Introduction to Abstract Algebra II: Homework 8

Due on March 11, 2026 at 23:59

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Exercise 35.16. Let D be a Euclidean domain and let ν be a Euclidean norm on D. Show that for nonzero $a, b \in D$, one has $\nu(a) < \nu(ab)$ if and only if b is not a unit of D. [Hint: Argue from Exercise 15 that $\nu(a) < \nu(ab)$ implies that b is not a unit of D. Using the Euclidean algorithm, show that $\nu(a) = \nu(ab)$ implies $\langle a \rangle = \langle ab \rangle$. Conclude that if b is not a unit, then $\nu(a) < \nu(ab)$.]

Solution. Suppose that $\nu(a) < \nu(ab)$. We show that b is not a unit. Assume for contradiction that b is a unit. Then there exists $b^{-1} \in D$ such that $bb^{-1} = 1$. Multiplying ab by b^{-1} gives $a = (ab)b^{-1}$. Since $b^{-1} \in D$, this shows that a is a multiple of ab. In a Euclidean domain, the Euclidean norm is compatible with divisibility in the sense that if $x \mid y$ and $y \neq 0$, then $\nu(x) \leq \nu(y)$. Thus from $ab \mid a$ we obtain $\nu(ab) \leq \nu(a)$, which contradicts the assumption $\nu(a) < \nu(ab)$. Therefore b cannot be a unit.

Now suppose that b is not a unit. We show that $\nu(a) < \nu(ab)$. Assume for contradiction that $\nu(ab) \leq \nu(a)$. Since $a \neq 0$, apply the Euclidean division algorithm to divide a by ab. Then there exist $q, r \in D$ such that $a = q(ab) + r$, where either $r = 0$ or $\nu(r) < \nu(ab)$. If $r \neq 0$, then $\nu(r) < \nu(ab) \leq \nu(a)$. But from the equation $r = a - qab = a(1 - qb)$, we see that r is a multiple of a, so $a \mid r$. This implies $\nu(a) \leq \nu(r)$, contradicting $\nu(r) < \nu(a)$. Hence $r = 0$. Thus

$$a = q(ab) = (qa)b.$$

Since $a \neq 0$, cancelling a gives $1 = qb$, so b is a unit, contradicting the assumption. Therefore $\nu(ab) \leq \nu(a)$ is impossible, and we must have $\nu(a) < \nu(ab)$.

Thus, $\nu(a) < \nu(ab)$ if and only if b is not a unit in D. □

Exercise 35.17. Prove or disprove the following statement: If ν is a Euclidean norm on Euclidean domain D, then $\{a \in D | \nu(a) > \nu(1)\} \cup \{0\}$ is an ideal of D.

Solution. Let $D = \mathbb{Z}$, which is a Euclidean domain with Euclidean norm $\nu(n) = |n|$. Then $\nu(1) = 1$. The set in the statement becomes

$$S = \{a \in \mathbb{Z} \mid |a| > 1\} \cup \{0\}.$$

If S were an ideal of $\mathbb{Z}$, it would have to be closed under addition. However, this is not the case. For example, $2 \in S$, $-3 \in S$, yet $2 + (-3) = -1$. But $|-1| = 1$, so $-1 \notin S$. Thus S is not closed under addition. Since an ideal must be closed under addition, S is not an ideal of $\mathbb{Z}$.

Thus, the statement is *false* in general. □

Exercise 35.18. Show that every field is a Euclidean domain.

Solution. Let F be a field. We show that F is a Euclidean domain. Define a function $\nu : F \setminus \{0\} \to \mathbb{N}$ by

$$\nu(a) = 1 \quad \text{for all } a \in F \setminus \{0\}.$$

We verify the Euclidean property. Let $a, b \in F$ with $b \neq 0$. Since F is a field, b has a multiplicative inverse b^{-1}. Define

$$q = ab^{-1}, \qquad r = 0.$$

Then $a = bq + r$. Because $r = 0$, the Euclidean condition is satisfied: either $r = 0$ or $\nu(r) < \nu(b)$. Hence the required division property holds for all $a, b \in F$ with $b \neq 0$.

Thus, F is a Euclidean domain. □

Exercise 35.20. Let D be a UFD. An element c in D is a least common multiple (abbreviated lcm) of two elements a and b in D if $a \mid c$, $b \mid c$ and if c divides every element of D that is divisible by both a and b. Show that every two nonzero elements a and b of a Euclidean domain D have an lcm in D. [Hint: Show that all common multiples, in the obvious sense, of both a and b form an ideal of D.]

Solution. Let D be a Euclidean domain and let $a, b \in D$ be nonzero. Consider the set

$$I = \{x \in D : a \mid x \quad b \mid x\},$$

that is, the set of all common multiples of a and b. We first show that I is an ideal.

First note that $0 \in I$ since $a \mid 0$ and $b \mid 0$. If $x, y \in I$, then $a \mid x$ and $a \mid y$, so $a \mid (x - y)$. Similarly, $b \mid (x - y)$. Hence $x - y \in I$. If $r \in D$ and $x \in I$, then $x = ak_1 = bk_2$ for some $k_1, k_2 \in D$, so

$$rx = a(rk_1) = b(rk_2),$$

which implies $rx \in I$. Thus I is an ideal of D.

Since D is a Euclidean domain, every ideal of D is principal. Hence there exists $c \in D$ such that $I = \langle c \rangle$. Since $c \in I$, we have $a \mid c$ and $b \mid c$, meaning c is a common multiple of a and b.

Lastly, we show that c divides every common multiple of a and b. Let $x \in D$ such that $a \mid x$ and $b \mid x$. Then, $x \in I$. Since $I = \langle c \rangle$, then $x = cd$, for some $d \in D$. This implies that c also divides x. ☐

Exercise 36.6. Consider $\alpha = 7 + 2i$ and $\beta = 3 - 4i$ in $\mathbb{Z}[i]$. Find σ and ρ in $\mathbb{Z}[i]$ such that

$$\alpha = \beta\sigma + \rho \quad \text{with} \quad N(\rho) < N(\beta).$$

[Hint: Use the construction in the proof of Theorem 36.4.]

Solution. Dividing α and β, we get

$$\frac{\alpha}{\beta} = \frac{\alpha\bar{\beta}}{\beta\bar{\beta}} = \frac{(7 + 2i)(3 + 4i)}{(3 - 4i)(3 + 4i)} = \frac{13 + 34i}{25} = 0.52 + 1.36i.$$

Rounding 0.52 to 1, and 1.36 to 1, we get

$$\frac{\alpha}{\beta} = 1 + i = \sigma.$$

Now, let's compute the remainder

$$7 + 2i = (3 - 4i)(1 + i) + \rho \Rightarrow 7 + 2i = 7 - i + \rho \Rightarrow \rho = 3i.$$

Let's check that $N(\rho) < N(\beta)$. Computing each norm, we get $N(\rho) = 9 < 25 = N(\beta)$. Thus, we have

$$7 + 2i = (3 - 4i)(1 + i) + 3i.$$

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Exercise 36.15. Let $\langle a \rangle$ be a nonzero principal ideal in $\mathbb{Z}[i]$.

(i) Show that $\mathbb{Z}[i]/\langle a \rangle$ is a finite ring. [Hint: Use the division algorithm.]

(ii) Show that if σ is an irreducible of $\mathbb{Z}[i]$, then $\mathbb{Z}[i]/\langle \sigma \rangle$ is a field.

(iii) Referring to part (ii), find the order and characteristic of each of the following fields.

(a) $\mathbb{Z}[i]/\langle 3 \rangle$ (b) $\mathbb{Z}[i]/\langle 1 + i \rangle$ (c) $\mathbb{Z}[i]/\langle 1 + 2i \rangle$

Solution to (i). Let $a \in \mathbb{Z}[i]$ be nonzero. Recall that $\mathbb{Z}[i]$ is a Euclidean domain with Euclidean norm

$$N(x + yi) = x^2 + y^2.$$

By the division algorithm in $\mathbb{Z}[i]$, for every $\alpha \in \mathbb{Z}[i]$ there exist $q, r \in \mathbb{Z}[i]$ such that $\alpha = qa + r$, with either $r = 0$ or $N(r) < N(a)$. Thus every coset of $\mathbb{Z}[i]/\langle a \rangle$ contains a representative $r \in \mathbb{Z}[i]$ satisfying $N(r) < N(a)$. Therefore

$$\mathbb{Z}[i]/\langle a \rangle = \{r + \langle a \rangle \mid r \in \mathbb{Z}[i],\ N(r) < N(a)\}.$$

But there are only finitely many Gaussian integers $r = x + yi$ such that

$$x^2 + y^2 < N(a),$$

since this inequality bounds both x and y. Hence there are only finitely many possible representatives r, so $\mathbb{Z}[i]/\langle a\rangle$ has finitely many elements. Therefore $\mathbb{Z}[i]/\langle a\rangle$ is a finite ring. $\square$

Solution to (ii). Let σ be an irreducible element of $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a Euclidean domain, it is also a principal ideal domain, and in a PID every irreducible element generates a maximal ideal.

Thus $\langle\sigma\rangle$ is a maximal ideal of $\mathbb{Z}[i]$. A standard result from ring theory states that if M is a maximal ideal of a ring R, then the quotient ring R/M is a field. Therefore $\mathbb{Z}[i]/\langle\sigma\rangle$, is a field. $\square$

Solution to (iii). For $\mathbb{Z}[i]/\langle 3\rangle$, we have $N(3) = 9$. Thus, the quotient has 9 elements. Since $3 = (1+i)(1-i)$ in $\mathbb{Z}[i]$, the ideal $\langle 3\rangle$ is not generated by an irreducible, so the quotient is not an integral domain. However, its additive structure still has 9 elements. The characteristic is 3, since $3(1 + \langle 3\rangle) = 0$. Thus, $|\mathbb{Z}[i]/\langle 3\rangle| = 9$ and the characteristic is 3.

For $\mathbb{Z}[i]/\langle 1+i\rangle$, we have $N(1+i) = 2$. Thus the quotient has 2 elements. A field with 2 elements is $\mathbb{F}_2$, so $|\mathbb{Z}[i]/\langle 1+i\rangle| = 2$ and the characteristic is 2.

For $\mathbb{Z}[i]/\langle 1+2i\rangle$, we have $N(1+2i) = 5$. Since 5 is prime in $\mathbb{Z}$, $1+2i$ is irreducible in $\mathbb{Z}[i]$, so the quotient is a field with 5 elements. Thus $|\mathbb{Z}[i]/\langle 1+2i\rangle| = 5$, and the characteristic is 5. $\square$

Exercise 19.1. Find a basis $\{(a_1, a_2, a_3), (b_1, b_2, b_3), (c_1, c_2, c_3)\}$ for $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ with all $a_i \neq 0$, all $b_i \neq 0$, and all $c_i \neq 0$. (Many answers are possible.)

Solution. Consider the vectors

$$a = (1, 1, 1), \qquad b = (1, 1, 2), \quad \text{and} \quad c = (1, 2, 1).$$

All coordinates of these vectors are nonzero. To check that they form a basis for $\mathbb{Z}^3$, we compute the determinant of the matrix whose rows are these vectors:

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = -1.$$

Since the determinant is ± 1, the matrix is unimodular, and the vectors form a basis for $\mathbb{Z}^3$. Therefore

$$\{(1, 1, 1), (1, 1, 2), (1, 2, 1)\},$$

is a basis for $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ with all coordinates nonzero. $\square$

Exercise 19.4. Find conditions on $a, b, c, d \in \mathbb{Z}$ for $\{(a, b), (c, d)\}$ to be a basis for $\mathbb{Z} \times \mathbb{Z}$. [Hint: Solve $x(a, b) + y(c, d) = (e, f)$ in $\mathbb{R}$, and see when the x and y lie in $\mathbb{Z}$.]

Solution. Let $(a, b), (c, d) \in \mathbb{Z} \times \mathbb{Z}$. We want to determine when $\{(a, b), (c, d)\}$ is a basis for $\mathbb{Z} \times \mathbb{Z}$. A basis for $\mathbb{Z}^2$ means that every $(e, f) \in \mathbb{Z}^2$ can be written uniquely as

$$(e, f) = x(a, b) + y(c, d),$$

with $x, y \in \mathbb{Z}$. First solve the equation over $\mathbb{R}$. Writing components gives the linear system

$$\begin{cases} ax + cy = e, \\ bx + dy = f. \end{cases}$$

In matrix form,

$$\begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} e \\ f \end{pmatrix}.$$

Assume the determinant $\Delta = ad - bc$ is nonzero. Then the system has the unique real solution

$$x = \frac{ed - cf}{ad - bc} \quad \text{and} \quad y = \frac{af - be}{ad - bc}.$$

For $(a, b), (c, d)$ to be a basis of $\mathbb{Z}^2$, these expressions must lie in $\mathbb{Z}$ for every $e, f \in \mathbb{Z}$. This happens exactly when

$$ad - bc = \pm 1.$$

Indeed, if $ad - bc = \pm 1$, then the denominators are ± 1, so x and y are always integers. Conversely, if $|ad - bc| > 1$, choose $(e, f) = (1, 0)$. Then

$$x = \frac{d}{ad - bc} \quad \text{and} \quad y = \frac{-b}{ad - bc},$$

which cannot both be integers unless $|ad - bc| = 1$. Hence not every element of $\mathbb{Z}^2$ can be written as an integer combination. Therefore $\{(a, b), (c, d)\}$ is a basis for $\mathbb{Z} \times \mathbb{Z}$ if and only if

$$ad - bc = \pm 1. \qquad \square$$

Exercise 19.10. Show that a free abelian group contains no nonzero elements of finite order.

Solution. Let G be a free abelian group with basis $\{e_i\}_{i \in I}$. By definition, every element $g \in G$ can be written uniquely as a finite linear combination

$$g = n_1 e_{i_1} + n_2 e_{i_2} + \cdots + n_k e_{i_k},$$

where $n_1, \ldots, n_k \in \mathbb{Z}$ and the e_{i_j} are distinct basis elements.

Suppose that $g \in G$ has finite order. Then there exists a positive integer m such that $mg = 0$. Substituting the expression for g, we obtain

$$m(n_1 e_{i_1} + n_2 e_{i_2} + \cdots + n_k e_{i_k}) = (mn_1) e_{i_1} + (mn_2) e_{i_2} + \cdots + (mn_k) e_{i_k} = 0$$

Since the representation of elements in terms of the basis is unique, the coefficients of each basis element must be zero. Thus

$$mn_1 = mn_2 = \cdots = mn_k = 0.$$

Because $\mathbb{Z}$ has no nonzero elements of finite order, it follows that $n_1 = n_2 = \cdots = n_k = 0$. Hence $g = 0$. Therefore the only element of G with finite order is the identity element. Thus a free abelian group contains no nonzero elements of finite order. $\square$